

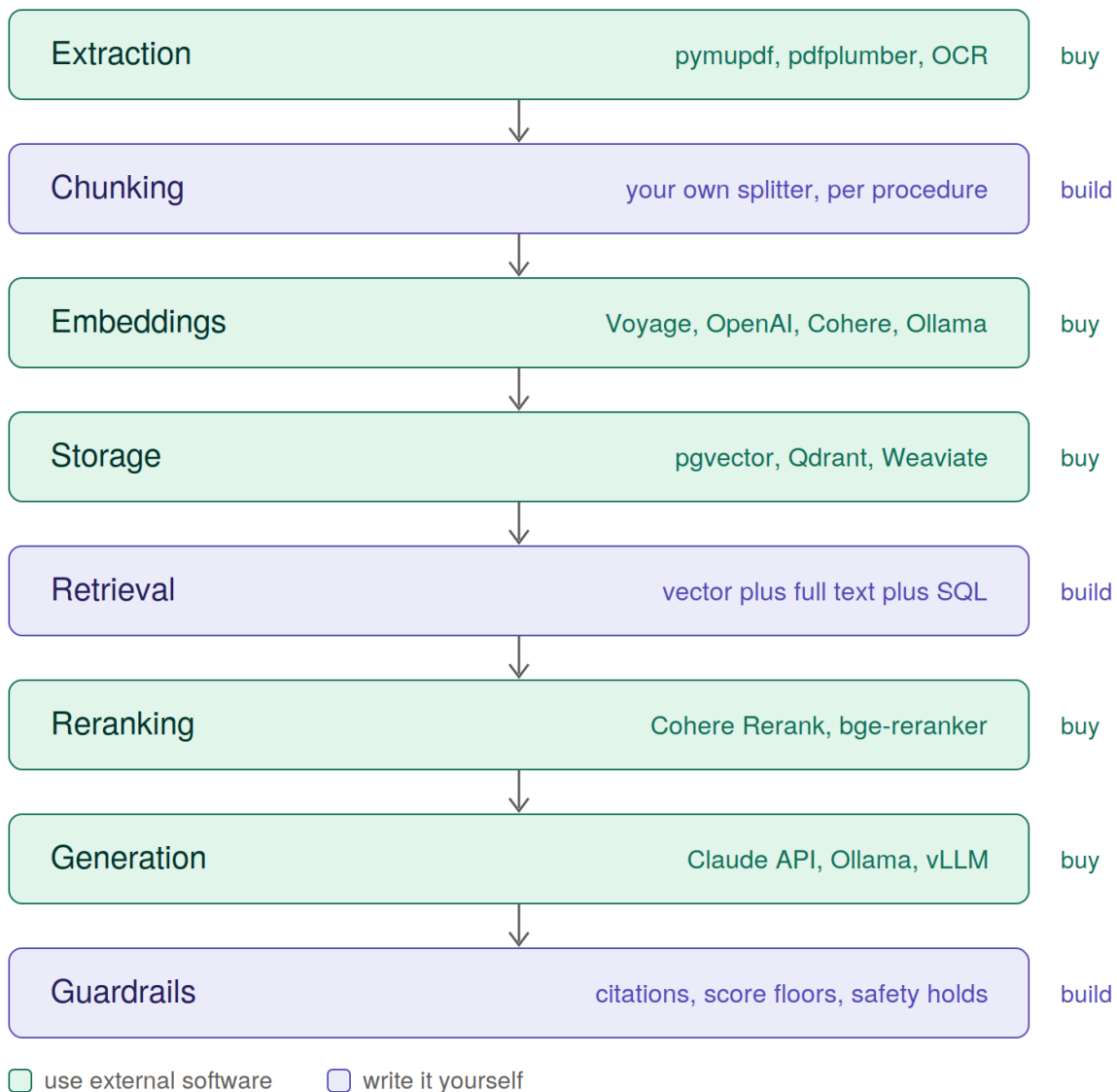
Maintenance LLM interview cheat sheet

2026-09-20 · @Someone

Talking points for a maintenance or reliability AI role: the stack, the three layers worth building, the numbers, and the traps.

The stack, eight layers

Five layers are commodity and you buy them. Three carry your value and you write them.



A PDF enters at the top and an answer leaves at the bottom.

Layer	Build or buy	Options	What it decides
Extraction	Buy	pymupdf, pdfplumber, Tesseract for scans	Whether the text exists at all
Chunking	Build	Your own splitter, per procedure or fault code	What can be found
Embeddings	Buy	Voyage, OpenAI, Cohere, or bge-m3 and nomic via Ollama	Match quality, and it locks your schema
Storage	Buy	pgvector in Postgres; Qdrant, Weaviate, Milvus past roughly 10M vectors	Whether you can join vectors to plant data
Retrieval	Build	Vector plus full text plus SQL joins	What actually surfaces
Reranking	Buy	Cohere Rerank, bge-reranker-v2-m3 locally	Cheapest quality win available
Generation	Buy	Claude API, or Llama and Qwen via Ollama or vLLM	Answer phrasing, not answer correctness
Guardrails	Build	Citations required, score floors, safety holds	Whether a tech trusts it at 2am

The line to remember: about 80 percent of the system is off the shelf, and the 20 percent you write decides whether it works.

The three layers that carry your value

Every vendor ships the same five bought layers. These three are where a plant system beats a generic one, and they are what a hiring manager is really probing.

Chunking. Generic chunkers split on character counts. Maintenance manuals split on procedures, fault codes and torque tables. A splitter that treats an alarm code section as one unit beats any library default, and it is about 100 lines of code. Say: the unit of retrieval should match the unit a technician asks about.

Retrieval. Vector search is bad at exact identifiers. When a tech types E633 you want a literal lookup, not a nearest neighbor guess. Postgres gives you full text search beside pgvector, so run both, merge the results, then rerank. Say: hybrid, because plant questions mix symptoms with part numbers.

Guardrails. Require a page citation on every answer. Refuse to answer when no chunk clears a score threshold. Flag lockout, tagout, pressure and electrical procedures for human confirmation rather than answering them. Say: in a plant, a confident wrong answer is worse than no answer.

Numbers and terms to have ready

Term	What to say
RAG	Retrieval augmented generation. Retrieve relevant text, then let the model answer from it instead of from memory
Embedding	A vector, a few hundred to a few thousand numbers, where similar meaning lands close together
Dimension	Fixed by the model and by your column. nomic 768, bge-m3 and voyage-3 1024, OpenAI large 3072. Switching models means re-embedding everything
Chunk size	Typically 300 to 500 words, split on natural boundaries, with a sentence or two of overlap
Hybrid search	Vector similarity plus keyword or full text, merged. Needed because exact codes fail vector search
Reranking	A cross encoder reorders the top 50 down to the best 5. Biggest quality gain per dollar
Top k	How many chunks you feed the model. Retrieve wide, rerank, pass few
Chunk metadata	Document, page, asset, part. This is what turns an answer into a citation

Term	What to say
Eval	Write 15 to 20 real technician questions, check whether the right chunk lands in the top 3. Ragas or TruLens if you want it automated
Cost shape	Embeddings cost cents once. Reranking runs around a dollar per thousand queries. Generation dominates the bill

One caution worth stating out loud: RAG systems fail silently. Without an eval set you are judging by vibes.

Questions they ask, and the demo

Why not just buy MaintainX or Limble? You should, for work orders, mobile and procurement. Build the retrieval layer when your data model is specific to the plant, because that is the part a vendor will not let you extend.

How do you stop it hallucinating? Retrieval grounds the answer, a score floor refuses when nothing relevant comes back, and every answer carries a page citation so the tech can verify in ten seconds.

How do you keep it current? Delete the document row and let the cascade clear its chunks, then reingest. Record the embedding model on the document so you know what needs regenerating if you switch.

On prem or cloud? Embeddings and retrieval run fine on prem. The question is whether manuals and work history may leave the network, which is a policy answer, not a technical one.

How do you measure it? Retrieval recall at 3 on a question set written by technicians, not by me. Then time to answer, and how often the tech opened the cited page and found it right.

What can you show? Postgres 17 with pgvector in Docker, an asset to part to document to chunk schema, 2000 alarm records, Grafana dashboards, and an MCP server letting Claude query the database in natural language with read only credentials. Built on a home machine, not a slide.

Traps, and the honest answer

Trap	What to say
Tag quality	Historian alarms usually join on a text tag with no foreign key. Renamed or mistyped tags break the join silently. Tag cleanup comes before any AI work
Generic alarm codes	The same code fires on different machines and implies different parts. You need a mapping from code plus asset to likely parts, and only plant people can build it
Scanned drawings	No text layer means nothing to embed. OCR first, and accept that dense schematics retrieve poorly whatever you do
Tables in manuals	Torque specs shredded into prose become noise. Put structured data in real tables and skip embedding it
Safety procedures	Never let a generated answer stand in for a lockout or tagout procedure. Route to the document, do not paraphrase it
Overpromising predictive	Predicting failure needs condition data, not manuals. Different problem, different sensors, different budget
Adoption	A tech who gets one wrong answer stops using it. Citations and refusals matter more than fluency

If asked what you would not do: I would not train a model. Fine tuning on plant documents is expensive, goes stale on the next manual revision, and loses the citation trail that makes an answer checkable.